

Question 1: Radix Sort (Radix = 9)

Array: {179, 721, 639, 549, 292, 427, 335, 435, 62}

Pass 1 — Ones digit (digit % 9):

Bucket	Elements
0	179,639,549
1	721
2	292,62
5	335,435
7	427

{179, 639, 549, 721, 292, 62, 335, 435, 427}

Bucket	Elements
0	292
2	721,427
3	639, 335, 435
4	549
6	62
7	179

{292, 721, 427, 639, 335, 435, 549, 62, 179}

Bucket	Elements
0	62
1	179
2	292
3	335
4	427,435
5	549
6	639
7	721

{62, 179, 292, 335, 427, 435, 549, 639, 721}

Question 2: Sort 4 Elements in 5 Comparisons

Lower bound check: $\lceil \log_2(4!) \rceil = \lceil \log_2(24) \rceil = 5$.

Algorithm (elements A, B, C, D):

1. Compare A vs B so $A < B$
2. Compare C vs D so $C < D$
3. Compare B vs D so $B < D$ now we know $A < B < D$ and $C < D$
4. Compare C vs B:
 - a. $C < B$: C sits below B. Now insert C relative to A:
 - i. Step 5: Compare C vs A $\rightarrow$ result is either $A < C < B < D$ or $C < A < B < D$
 - b. $C \geq B$: Then $B < C < D$. Since $A < B$, we already know $A < B < C < D$ — **done in 4 steps**

Worst case = **exactly 5 comparisons**, matching the lower bound.

Question 3: FBSSort Algorithm

Using 1-based indexing

Algorithm:

FBSSort(A, n):

1. Sort A in ascending order
2. Split into:
 - small = first $\lfloor n/2 \rfloor$ elements go to odd positions //ascending, already sorted
 - large = remaining $\lfloor n/2 \rfloor$ elements go to even positions //place in reverse for descending
3. Reconstruct array by interleaving

Verification on {7,20,10,19,10,17,14,15,15}:

- Sorted: {7,10,10,14,15,15,17,19,20}
- small = {7,10,10,14,15} $\rightarrow$ positions 1,3,5,7,9 // ascending
- large = {15,17,19,20} reversed $\rightarrow$ {20,19,17,15} // positions 2,4,6,8 descending ✓

- All even-position values $\geq$ all odd-position values

Running time: $O(n \log n)$; the split and placement are $O(n)$.